

Mathematical Frameworks — Module 08: Planning —

Study Questions

1. How does Active Inference treat planning? (as inference over policies)
2. What is the policy prior $P(\pi)$? What does it encode?
3. Write the posterior over policies: $P(\pi | o) \propto \dots$
4. How does the agent evaluate each policy? (by computing EFE)
5. What is a deep temporal model? How does it handle multiple timescales?
6. In a hierarchical POMDP, what does the higher level set for the lower level?
7. How does habit formation emerge mathematically? What happens to $P(\pi)$?
8. What is the difference between goal-directed and habitual behavior in this framework?
9. When $P(\pi)$ is very strong, what happens to EFE evaluation?
10. List the 7-8 steps of the complete Active Inference algorithm.
11. How does tree search work in Active Inference? Trace a 3-step policy evaluation.
12. What is the role of γ (policy precision) in the planning process?
13. How does model-based planning differ from model-free habitual action?
14. How does sleep (BMR) contribute to planning?
15. Can an Active Inference agent plan infinitely far into the future? What limits planning depth?
16. How does the hierarchical structure of deep temporal models relate to the cortical hierarchy?
17. What is the computational cost of evaluating all policies? How is this managed?
18. How does Active Inference planning differ from classical AI planning (A^* , STRIPS)?
19. What happens mathematically when an agent encounters a novel situation where habits don't apply?
20. Reflecting on the entire Mathematical Frameworks unit: How do all 8 modules connect to form a complete theory?